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An information-theoretic argument for the restriction of the current biological alphabets to 4 nucleotides and 20 amino acids

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Life as we know it is based on foldable biopolymers encoded with just 4 nucleotides or 20 amino acids. Evolution of these biopolymers requires effective and fast search of both the conformational space for folding and the sequence space for evolution. Energy landscape theory links the free energy of the possible polymer sequences and conformations with its ability to fold, while molecular information theory provides connections between conformational entropy and sequence entropy. Combining these two theories provides constraints to the alphabet size of an evolving biopolymer, given its physico-chemical properties. Empirical estimations of the size of the effective sequence and conformational spaces of foldable proteins and RNA show that the observed alphabet sizes agree with the theoretical predictions and are just large enough to allow for biopolymer evolution. In this scenario, one effective digital alphabet letter in the sequence landscape maps to one effective analog monomer configuration in the conformational landscape. We also use the current views on genetic code evolution to explore scenarios for biopolymers in the early stages of life. We find that primitive genetic systems coding for smaller amino acid alphabets may have led to a prominent presence of intrinsically disordered proteins.

Keywords Biopolymer, Random energy model, Genetic code

Terrestrial biochemistry is based on the existence of linear polymers of just a handful of monomers. A fundamental aspect of Biology is that the functional structures of these molecules can be *coded* in linear strings of units, many of which, in the appropriate conditions, rapidly and spontaneously fold into stable structures¹. In turn, the exploration of the vast sequence spaces that encodes structures has to be efficient enough such that evolution can proceed¹. It has long been debated why are current polypeptides and polynucleotides encoded with just 20 and 4 monomers respectively, out of the many chemically plausible alternatives^{2,3}. Here we show that the conditions for fast folding and effective evolution strongly constraint the coding alphabet sizes to just the currently used by proteins and RNA.

A simple theory for biopolymer alphabet size

Energy landscape theory relates the free energy of the possible conformations and sequences of a polymer with the ability to find sequences compatible with a foldable biologically relevant structure¹. Molecular information theory emphasizes the relationship between conformational entropy and sequence entropy⁴. We have previously integrated molecular information theory with energy landscape theory in a manner that allows us to examine the role alphabet size in biopolymer folding and evolution^{4,5}.

From the viewpoint of molecular information theory, spontaneous folding is a self-recognition molecular process where information is gained by finding the native conformation among the non-native alternatives. This gain in information (or decrease in conformational entropy) during folding, $R_{Levinthal}$, can be defined per protein site (or amino acid) and folding operation (folding event). $R_{Levinthal}$ is a quantification of the entropic challenge in Levinthal's paradox and is related to the effective number of conformations per protein site in the unfolded

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and folded states ($N_{unfolding}$ and N_{native}) (Eq. 1, left)⁵. The effective number of conformations accessible to an amino acid in the unfolded state that are relevant to the folding search can be evaluated using measurements of backbone conformational entropy, secondary structure alphabets, and from structure prediction methods from residue burial layers and from residue contacts⁵. Empirical estimations reveal an average value of $R_{Levinthal}$ of 2.4 ± 0.3 bits/(site-operation) for natural proteins⁵. $R_{Levinthal}$ corresponds to the loss in entropy upon formation of the structural features of the native state that are sufficient to infer the remaining details of the folded structure⁵. Since these features are a subset of the total entropy change, $R_{Levinthal}$ is lower than the loss in conformational entropy upon folding⁶.

Molecular information theory also postulates that the information gained by finding the native state by folding matches the information gained (the decrease in sequence entropy) upon going from a random sequence to the ensemble of evolved sequences that are compatible with a given structure, $R_{sequence}$ ⁵. $R_{sequence}$ can be related to the size of the monomer alphabet and the effective number of monomers per site in the ensemble of evolved sequences (*Alphabet size* and $N_{evolved}$) (Eq. 1, right)⁵. Empirical estimations based on extant protein sequences show that $R_{sequence}$ is on average 2.1 ± 0.3 bits/(site-operation), in agreement with the average value of $R_{Levinthal}$ and with the theoretical prediction⁵.

$$R_{Levinthal} = \log_2 \left(\frac{N_{unfolding}}{N_{native}} \right) = R_{sequence} = \log_2 \left(\frac{\text{Alphabet size}}{N_{evolved}} \right) \quad (1)$$

The equality in Eq. 1 implies the following relationship between *Alphabet size*, $N_{evolved}$, $N_{unfolding}$ and N_{native} :

$$\frac{\text{Alphabet size}}{N_{evolved}} = \frac{N_{unfolding}}{N_{native}} \quad (2)$$

In other words, alphabet size is not free to take any value in an evolved biopolymer, but linked to the values of $N_{evolved}$, $N_{unfolding}$ and N_{native} .

We may apply energy landscape theory to a random energy model to identify the conditions in which it is feasible to find sequences that fold into a given structure^{5,6}. There are viable sequences if the entropy in sequence space for a given target structure is positive. This is the case if the number of monomers per site of an evolved sequence ($N_{evolved}$) is larger than the effective number of conformations per site in the unfolded biopolymer^{6,7} ($N_{unfolding}$):

$$N_{evolved} > N_{unfolding} \quad (3)$$

Combining Eq. [1] and Eq. [2] at the limit of $N_{evolved} = N_{unfolding}$, it follows that

$$(N_{unfolding})^2 = \text{Alphabet size} \cdot N_{native} = (N_{evolved})^2 \quad (4)$$

That is, the application of a key energy landscape theory result to Eq. [2] leads to the conclusion that the product of the alphabet size and N_{native} can be understood as constrained by either or both $N_{unfolding}$ or $N_{evolved}$.

Extant folded biopolymers can be usually well described using a single reference structure and small fluctuations around it. In this case, we can approximate N_{native} to 1 and

$$(N_{unfolding})^2 = \text{Alphabet size} = (N_{evolved})^2 \quad (5)$$

Equation [2] constraints the ratio of *Alphabet size* and $N_{evolved}$, but not their absolute values. For a soluble polymer evolving in a given solvent, $N_{unfolding}$ is determined by the physico-chemistry of the polymer-solvent system. Given a value of $N_{unfolding}$, Eq. [5] fully constraints $N_{evolved}$ and *Alphabet size*. This result should be valid for any evolved and foldable biopolymer, terrestrial or not⁷, as long as energy landscape theory and molecular information theory are valid.

Combining Eqs. [1] and [5], it follows that the information content that an evolved polymer can encode is

$$R_{sequence} = \log_2 \left(\sqrt{\text{Alphabet size}} \right) \quad (6)$$

Which amounts to 1 bit/site for RNA and 2.26 bits/site for proteins.

Empirical test of the theory

The theoretical results pose three clear relationships between *Alphabet size*, $N_{unfolding}$ and $N_{evolved}$, namely that both $N_{unfolding}$ and $N_{evolved}$ should be equal to each other and to the square root of the *Alphabet size*. As the current *Alphabet size* is 20 amino acids in the case of proteins and 4 nucleotides in the case of RNA, Eq. [5] also predicts that $N_{unfolding} = N_{evolved} = 2$ for RNA and $N_{unfolding} = N_{evolved} = 20^{0.5} \approx 4.47$ for proteins (Fig. 1).

We previously approximated $N_{unfolding}$ of proteins from six estimations of the polypeptide backbone conformational entropy, the effective secondary structure alphabet, structure prediction from residue burial layers and structure prediction from residue-residue contacts⁵, obtaining an average value of 5.38 ± 0.46 effective conformations per amino acid. To evaluate $N_{unfolding}$ of RNA we used estimations of the loss in backbone conformational entropy upon RNA folding. Analysis of the 23 S ribosomal subunit yields a loss in backbone conformational entropy of 1.52 cal/(mol·K) per nucleotide⁸. A similar analysis of several ribozymes and riboswitches yields an average loss in backbone conformational entropy of 2.20 cal/(mol·K) per nucleotide⁹. Since $N_{unfolding} = e^{\frac{\Delta S}{R}}$, the estimated values for $N_{unfolding}$ are 2.13 and 3.00 effective conformations per

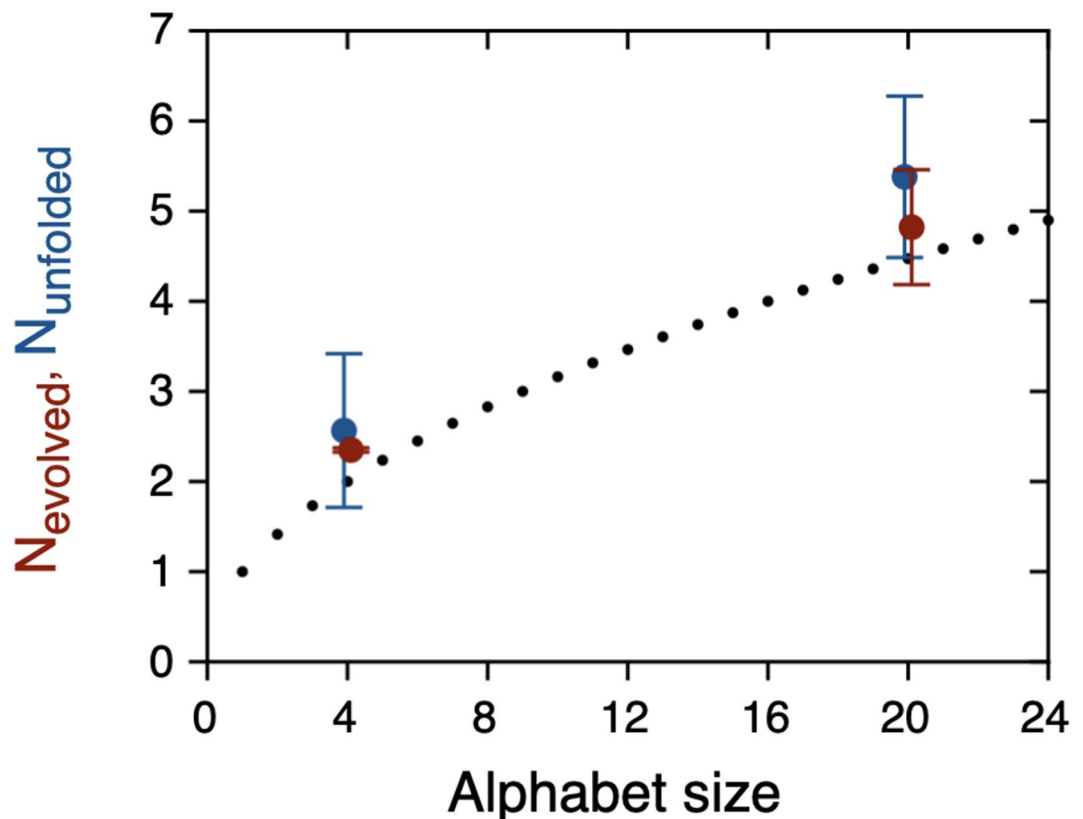


Fig. 1. Relationship between Alphabet size, $N_{evolved}$ and $N_{unfolded}$ for biopolymer folding and evolution. The black circles indicated the expected value for each Alphabet size according to Eq. [5]. The red circles indicate empirical estimates of $N_{evolved}$ for folded RNA and protein molecules. The blue circles indicate empirical estimates of $N_{unfolded}$ for folded RNA and protein molecules. The error bars indicate the mean plus/minus 1.96 times the standard error of the mean (95% confidence interval for estimation of the mean value). Red and blue circles are slightly displaced along the x-axis for clarity.

nucleotide, respectively. We also approached RNA folding as the finding of one base pair among all possible pairs for each nucleotide of the chain. As a first approximation, we consider that the searches in the different sites in an RNA chain are independent. We can calculate the effective size of the base pair structural alphabet for RNA using the base pair abundances observed in natural RNA structures. Performing this calculation for the data in Table 3 of¹⁰ yields a value for $N_{unfolded}$ of 2.48 effective conformations per nucleotide. From our three estimates of $N_{unfolded}$, we calculated an average $N_{unfolded}$ for RNA of 2.54 ± 0.25 effective conformations per nucleotide. The values of $N_{unfolded}$ for both proteins and RNA are close to the square root of the respective Alphabet size, fulfilling the first prediction of the model (Fig. 1, blue circles).

We previously estimated the $N_{evolved}$ of proteins from an Alphabet size of 20 and seven estimations of the sequence entropy of natural proteins obtaining an average value of 4.82 ± 0.32 effective amino acids per site⁵. Taking into account the correlations between sites did not strongly affect this value⁵. Here, we estimate the $N_{evolved}$ of naturally occurring folded RNA molecules which can be calculated from sequence variations within natural RNA families. We used the methodology in⁵ to calculate the average $N_{evolved}$ over all sites with less than 50% gaps in over 1200 RNA alignments from the Rfam database¹¹, considering that the sequence restrictions at the different sites in an RNA chain are independent. The average value of $N_{evolved}$ over all alignments is 2.35 ± 0.024 effective nucleotides per site. The values of $N_{evolved}$ for both proteins and RNA are close to the square root of the respective Alphabet size, fulfilling the second prediction from the model (Fig. 1, red circles). Evolved protein and RNA sequences contain enough information to code for an efficient search of the conformational space during folding^{5,12}. The blue and red circles in Fig. 1 also show that the values of $N_{evolved}$ coincide with those for $N_{unfolded}$ for both proteins and RNA, fulfilling the third and last prediction of Eq. [5]. This ensures an efficient search of the sequence space in the evolution of foldable biopolymers⁵.

Implications for the evolution of biopolymers and their synthesis machinery

As discussed above, $N_{unfolded}$ and $R_{Levinthal}$ are determined by the chemical-physics of natural polymers. The evolution of folded biopolymers also requires that the Alphabet size of proteins and RNA is the square of the relevant $N_{unfolded}$. Equation [2] to [5] can be used to illustrate a plausible evolution of a foldable biopolymer, in the case of $N_{native} = 1$. Evolution may start from a sequence that is close to random, for which $N_{evolved}$ equals approximately Alphabet size and $R_{sequence}$ is close to 0. As $N_{evolved}$ decreases due to the constraints imposed by

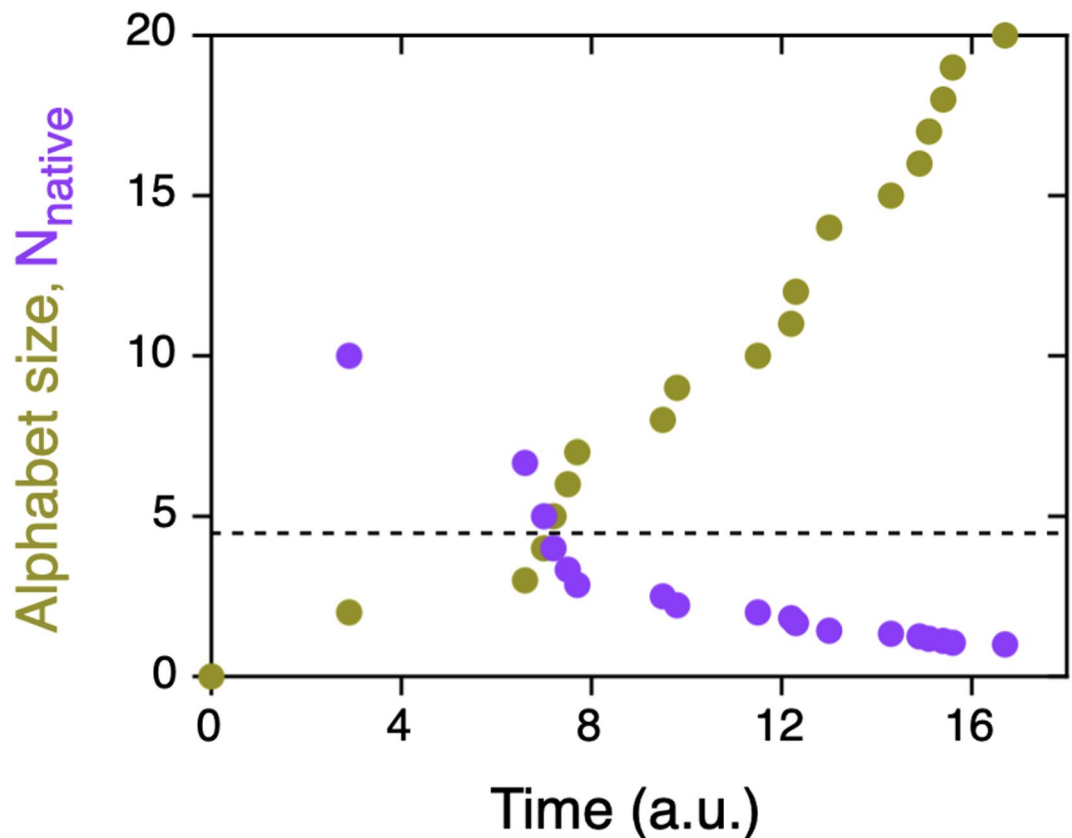


Fig. 2. A plausible scenario for the evolution of biopolymer alphabet size and conformational heterogeneity. Green circles: Alphabet size from¹⁴. Purple circles: N_{native} from Eq. [4]. The dashed line indicates the minimum value of Alphabet size that allows for an efficient search of folded protein sequences (approximately 4.47).

folding into a certain three-dimensional structure, R_{sequence} increases. When N_{evolved} reaches its limit value of N_{unfolded} , R_{sequence} matches the required value of $R_{\text{Levinthal}}$. The actual shape of the evolutionary sequence dynamics would depend on the corresponding energy landscape as illustrated for a simple case in¹³. Further decreases in N_{evolved} would have two effects. First, from the viewpoint of molecular information theory it would lead to excess information in evolved biopolymer sequences, which should evolve (dissipate) back to zero due to maximization of entropy. Second, from the viewpoint of energy landscape theory it would render the search for foldable sequences unfeasible due to N_{evolved} being lower than N_{unfolded} . We thus expect from theory that the average N_{evolved} of natural sequences is close to its limit value of N_{unfolded} . To sum up, Eq. [5] shows that biopolymer chemical-physics (i.e., N_{unfolded}) strongly constrains both *Alphabet size* and the diversity of evolved sequences (i.e., N_{evolved}).

We can also consider the evolution of biopolymers and their synthesis machinery in the early stages of terrestrial life using Eq. [4], which posits that the product $N_{\text{native}} \cdot \text{Alphabet size}$ is a constant. The minimum value of N_{native} is 1, as discussed above for extant biopolymers. In that case, *Alphabet size* reaches its maximum value of $(N_{\text{unfolded}})^2 = (N_{\text{evolved}})^2$. The minimum possible value for *Alphabet size* to fulfill Eq. [3] is N_{unfolded} in which case N_{native} reaches a maximum value of $N_{\text{unfolded}} = N_{\text{evolved}}$. In other words,

$$N_{\text{native}} \in [1, N_{\text{evolved}} = N_{\text{unfolded}}]; \text{Alphabet size} \in [N_{\text{evolved}} = N_{\text{unfolded}}, (N_{\text{unfolded}})^2 = (N_{\text{evolved}})^2] \quad (7)$$

We can use Eq. [7] to explore a plausible scenario for the evolution of biopolymers early in the history of life. Given the evolutionary choice of a certain biopolymer chemistry, such as ribonucleotides or amino acids, N_{unfolded} is a constant. Trifonov and coworkers combined several lines of evidence to deduce the time course for incorporation of amino acids into the genetic code as a function of an ad hoc time scale¹⁴. Figure 2 uses this result to plot protein *Alphabet size* as a function of this semi quantitative time scale (green circles).

We observe that at early times *Alphabet size* takes values of 2 and 3, below the minimum necessary to evolve foldable proteins ($N_{\text{unfolded}} \approx 4.47$, dashed horizontal line), in agreement with previous proposals [15]. In this initial stage, biopolymer synthesis would not have efficiently led to folded molecules and may have contributed to organism fitness in a different manner. It has indeed been suggested that liquid-liquid phase separation of low complexity peptides may have provided structural support for membrane-less protocells [16].

Figure 2 also shows a quick rise of *Alphabet size* from 3 to 7 (glycine, alanine, valine, aspartic acid, glutamic acid, proline and serine) around an ad hoc time of 7.5. After this rapid transition, the present theory predicts that it would have been possible to evolve folded proteins since *Alphabet size* is larger than N_{unfolded} . At this stage, the mean value for N_{native} in evolved proteins would be around 3 conformations per monomer for proteins. This value is consistent with intrinsically disordered (yet not conformationally random) proteins. This is in agreement with the fact that five out of seven amino acids in this alphabet (glycine, aspartic acid, glutamic acid, proline and serine) have disorder-promoting propensities^{17,18}. In addition to formation of membraneless compartments via liquid-liquid phase separation, activities of early evolved proteins likely include RNA chaperones and ion channels^{17,18}.

In later stages of evolution (ad hoc times between 7.5 and 17) *Alphabet size* grew at a roughly constant rate as more amino acids were incorporated into the protein synthesis machinery. As the product $N_{\text{native}} \cdot \text{Alphabet size}$ is a constant (Eq. [4]), we hypothesize that N_{native} became smaller and the native states of the average protein progressively became better determined. This agrees with the incorporation of mainly order-promoting residues during this stage and may have allowed the appearance of proteins with specific catalytic activity^{17–19}. The current genetic code uses an alphabet of 20 amino acids. A well-folded protein, which has a single, stable structure (an N_{native} of 1) can be built from this alphabet. However, intrinsically disordered proteins demonstrate that this is not always the case, as they can be functional without adopting a single, fixed conformation.

Discussion

We present a simple theoretical result for the restriction of the *Alphabet size* of evolved foldable biopolymers. The potential for folding and evolution is fundamentally constrained by the physico-chemical degrees of freedom of the soluble polymer that determine N_{unfolded} . For spontaneous unassisted folding, the information contained in evolved sequences must be enough to *code* for the structures of the native folded states ($R_{\text{sequence}} = R_{\text{Levinthal}}$). This is mathematically equivalent to the quantification of their functional information²⁰. The minimum mapping between the conformational and the evolutionary landscapes is thus $N_{\text{evolved}} = N_{\text{unfolded}}$ such that one effective digital character is matched with one effective analog conformation. This directly bounds the minimum digital *Alphabet size*. The simplest codable and foldable polymer would be that one for which $N_{\text{unfolded}} \approx 2$, restricting the *Alphabet size* to 4 digits. Foldable RNA molecules appear to fall in this category. Detailed computational studies on the design of foldable RNA sequences coincide to show that an *Alphabet size* of 4 was the minimum required¹². We interpret that these results implicitly account for the role of metal ion coordination, kinetic traps and tertiary structure in RNA folding. Quantification of the relative importance of these factors and of (when present) catalytic constraints may require additional theoretical developments. For proteins, the polypeptide backbone geometry dictates that $N_{\text{unfolded}} = 5.38 \pm 0.46$, and as such *Alphabet size* must be larger than for RNA, in the order of 20 ~ 39. Since amino acids in a viable alphabet should all be sufficiently different from each other in terms of their physicochemical properties²¹, we interpret this number as 20 ~ 39 sufficiently different amino acids. The fact that the *Alphabet size* of modern genetically coded proteins is on the lower end of this range may reflect additional constraints on protein evolution, such as the size of the molecular systems involved in protein synthesis, the presence of post-translational modification systems, the multiple conformational substates involved in protein function and that further information besides folding needs to be coded for function^{22,23}. For proteins $N_{\text{evolved}} = 4.82 \pm 0.32$, which is in line with the theoretical and experimental findings for the minimal number of amino acids needed to specify a fold^{5,24}. The chemical nature of polynucleotide and polypeptide chains imply that a minimal non-overlapping genetic code relating them must be one of degenerate triplets. Our results suggest that the early evolution of the genetic code was accompanied by two transitions in the nature of the encoded proteins. First, low-complexity proteins were replaced by evolvable, yet intrinsically disordered proteins as the protein *Alphabet size* abruptly exceeded 5 amino acids. Second, the structures of many evolved proteins became progressively better determined as the protein *Alphabet size* gradually increased up to the current 20 genetically coded amino acids. The present theory can account for the fact that current intrinsically disordered proteins retain a significant amount of conformational entropy under native conditions. Furthermore, since N_{native} can be reduced to the limiting value of 1 with an *Alphabet size* of 20, natural incorporation of additional amino acids into the genetic code is not expected to take place. Thus, the evolution of the genetic code may be seen as largely complete.

Data availability

All data used/generated in this work are available from the corresponding authors (Diego U. Ferreira (ferreiro@qb.fcen.uba.ar), Ignacio Enrique Sánchez (isanchez@qb.fcen.uba.ar)) upon request.

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We call the attention of the international scientific community about the catastrophic erosion of Argentina's strong scientific tradition due to current funding constraints and the sudden termination of long term policies.

Author contributions

EAG DUF and IES designed research, performed research, analyzed data, and wrote the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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